

Total No. of Questions : 8]

SEAT No. :

PC-1741

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T.E. (Artificial Intelligence and Data Science)

ARTIFICIAL NEURAL NETWORK

(2019 Pattern) (Semester - II) (317531)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagram must be drawn wherever necessary.
- 4) Assume suitable data if necessary.

- Q1) a) Draw the state transition diagram for the three types of Boltzmann Machine neural network and comment on the nodes and their connections. [9]
- b) Train the hetero-associative memory network using outer products rule to store input row vectors $s = (s_1, s_2, s_3, s_4)$, to the output row vectors $t = (t_1, t_2)$. Use the vector pairs as given in Table 2. [9]

TABLE 2						
Input and targets	s_1	s_2	s_3	s_4	t_1	t_2
1 st	1	0	1	0	1	0
2 nd	1	0	0	1	1	0
3 rd	1	1	0	0	0	1
4 th	0	0	1	1	0	1

OR

- Q2) a) Discuss the False Minima problem, Stochastic Update and Simulated Annealing concepts with reference to the Boltzmann machines. [9]
- b) How do you define associative memory? What are the types of associative memories? Explain with 2 real applications of associative memories. [9]

P.T.O.

Q3) a) How Hebbian learning is different from competitive learning? What is winner- takes-it-all? Draw basic architecture of ART and discuss its principle of working. [9]

b) Discuss the following features of ART: [9]

- 1) self-scaling computational unit
- 2) self-adjusting memory search
- 3) Pre-learned pattern access and
- 4) Attention vigilance

OR

Q4) a) Construct and test an LVQ net with five vectors assigned to two classes. The given vectors along with the classes are as shown in Table given below. [10]

Vector	Class
[0 0 1 1]	1
[1 0 0 0]	2
[0 0 0 1]	2
[1 1 0 0]	1
[0 1 1 0]	1

b) Draw the architecture of Kohonen Network. Discuss the feature mapping concept. Explain the training algorithm of KSOFM. [8]

Q5) a) Draw comparative architecture of ANN and CNN. What is the need for CNN? Will the overfitting be less in CNN than ANN? And if yes, why does this happen? [9]

b) Illustrate (with diagram) the bias-variance dilemma? How the trade-off is achieved? Comment on: [8]

Bias Low with Variance High
Bias High with Variance Low

OR

Q6) a) Discuss the relevance and significance of [9]

- 1) pooling layer
- 2) padding
- 3) strided convolutions
- 4) dropout.

b) Compare and contrast: LeNet – 5, AlexNet, VGG –16 [8]

- Q7)** a) How neo-cognitron is different from MNIST handwritten recognition? Discuss in your own words with reference to the algorithmic steps. [9]
- b) How would you solve the problem of texture classification and segmentation? Develop your own block schematic for this task. [8]

OR

- Q8)** a) Develop your own NN model (algorithm) for Recognition of Olympic games symbols. Draw the architecture of your proposed system. Will you use backpropagation? Why? [9]
- b) How do we recognize consonant vowel (CV) segments? Is this signal different from Image classification project? Which NN algorithm would you use? [8]

